

From its inception, Ethereum has been developed openly, with a vibrant community of developers contributing to everything from the core protocol to thousands of decentralised applications (DApps) built on top of it.

The Ethereum Improvement Proposal (EIP) system mirrors Bitcoin's in its collaborative, public approach to change. The open nature of Ethereum's code has allowed for rapid iteration, the successful transition from proof-of-work to proof-of-stake (via Ethereum 2.0), and a thriving ecosystem of DeFi, NFTs, and DAOs—all of which benefit from open review and community governance.

Ethereum's transparency and openness have also helped it recover from major challenges, like the infamous DAO hack in 2016, where the community openly debated and ultimately agreed to a controversial chain split—leading to Ethereum and Ethereum Classic.

**Lessons from failed closed source blockchain projects:** On the other end of the spectrum, several blockchain projects that opted for closed source or opaque development have faced credibility issues, security risks, and community distrust. Projects like OneCoin and BitConnect serve as cautionary tales.

- OneCoin claimed to be a cryptocurrency but never had a publicly verifiable blockchain or open codebase. This lack of transparency helped it mask a multi-billion dollar Ponzi scheme that ultimately collapsed, leading to arrests and massive losses for investors.
- BitConnect, while initially appearing to be a legitimate platform, operated with a closed algorithm for its trading bot and reward system. Because no one could audit the code or verify how it worked, the platform collapsed amid fraud accusations once confidence waned.

These failures highlight a key lesson — when users can't verify

how a system operates, the risks of abuse, fraud, and collapse increase dramatically.

## Challenges and criticisms of open source blockchain

While open source is essential to blockchain transparency and decentralisation, it's not without its challenges. The very openness that fuels innovation and trust can also introduce risks—from security vulnerabilities to tensions around commercialisation. Understanding these issues is critical for anyone working within or evaluating blockchain ecosystems.

### **Security vulnerabilities and**

**risks:** Open source code is available for anyone to inspect—including hackers. While transparency allows for community-led audits and faster discovery of bugs, it also means that malicious actors can study the code to find and exploit vulnerabilities.

In blockchain, even small bugs can have massive consequences. A single coding flaw in a smart contract can be exploited to drain millions in assets, as seen in various decentralised finance (DeFi) hacks and exploits. Without strong, continuous code review and formal verification processes, open source projects may remain vulnerable despite their transparency. Another risk lies in dependency management. Many open source blockchain platforms rely on third-party libraries and tools that may themselves contain vulnerabilities. If not properly maintained or reviewed, these dependencies can become attack vectors.

### **Intellectual property and**

**commercialisation issues:** Open source encourages free access and modification of code—but this openness can create friction when it comes to monetisation, intellectual property (IP), and business models. Startups and enterprises building on open source blockchains sometimes struggle to protect proprietary innovations while still complying with open source licences.

There are also licence conflicts to consider. Different open source licences (like MIT, GPL, or Apache 2.0) come with varying requirements for reuse, redistribution, and attribution. Misunderstanding or misapplying these licences can lead to legal complications or disputes within the developer community.

Moreover, when core blockchain innovations are open source, it can be hard to maintain a competitive edge. Forking—where anyone can copy and modify the code to launch a competing network—is a common occurrence. This can lead to fragmentation, duplicate efforts, and confusion in the market.

## The future of blockchain transparency

As blockchain technology continues to evolve, so does the importance of transparency in securing user trust, regulatory compliance, and long-term sustainability. Open source remains the backbone of this transparency—but for it to keep pace with the growing complexity and impact of blockchain systems, deeper commitment and new strategies are required. The future lies in not just maintaining transparency, but strengthening it through broader adoption, community involvement, and supportive regulation.

### **Encouraging open source adoption:**

To foster greater trust and innovation, future blockchain projects—especially those involving financial services, public infrastructure, or governance—will need to embrace open source by default. Open source adoption should be seen not just as a best practice, but as a standard for legitimacy.

Encouraging this adoption will involve:

- Educating developers and startups on the benefits of open source for growth and credibility.
- Creating funding models (e.g., grants, DAOs, or foundation support) that sustain open source development.
- Recognising and rewarding contributors who improve open source blockchain ecosystems.